

Provenance Note

Real: torque spec (50-60 Nm for pipe/flange connections), the isolate-power-before-service and PPE safety language, and the suction-side failure modes (air/leakage in suction pipe, low inlet pressure) are paraphrased from real Grundfos CM/CMV centrifugal pump installation & operating manuals (net.grundfos.com, api.grundfos.com).

Real: the mechanical seal reference standard (API Standard 682, "Pumps - Shaft Sealing Systems for Centrifugal and Rotary Pumps") is a real, published API standard.

Synthetic: the specific operating pressure value, the exact 12-month replacement interval, the equipment tag "Pump-101", and the company/site context are fabricated for this demo -- no real manufacturer publishes a manual for equipment called "Pump-101".

Pump-101 Centrifugal Pump

OEM Installation, Operation & Maintenance Manual

Document No. OM-CP-101 | Revision 3 | Issued: 2024-02-10

This manual covers installation, operation, and routine maintenance of the Pump-101 series single-stage centrifugal pump, used in general industrial fluid-transfer service.

1. Technical Specifications

Parameter	Value
Rated operating pressure	12 bar
Maximum discharge pressure	14 bar
Rated flow rate	45 m ³ /h
Liquid temperature range	-10 C to 90 C
Seal type	Single mechanical seal, cartridge design, per API Standard 682
Recommended seal replacement interval	Every 12 months, or at first sign of leakage
Flange bolt torque (pipework connections)	50-60 Nm

Note: the pump must not be stressed by the connecting pipework. All flange connections must be tightened to the specified torque of 50-60 Nm, and pipes should be independently supported near the pump connection points.

2. Component List (Assembly)

Pump-101 is assembled from the following major components:

Component	Function
Impeller	Rotating element that imparts kinetic energy to the fluid
Mechanical seal (cartridge, API 682)	Prevents leakage along the rotating shaft
Bearing housing	Supports and locates the pump shaft
Coupling	Connects the pump shaft to the driving motor
Volute casing	Converts fluid velocity into pressure and directs discharge flow

3. Safety Precautions

Before performing any maintenance or servicing on Pump-101, the following precautions are mandatory:

- Switch off the electrical power supply to the pump motor and lock the main disconnect switch in the off position before starting any service work.
- Do not attempt to service the pump while it is pressurized or while the discharge/suction lines are still connected to an active process.
- Wear appropriate personal protective equipment (PPE) at all times, including safety glasses, gloves, and hearing protection where applicable.
- Verify that all rotating parts have come to a complete stop before removing any guards or covers.
- Confirm isolation of all energy sources (electrical, hydraulic, and stored pressure) before beginning disassembly.

4. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Pump does not deliver rated flow	Air in suction pipe or pump	Vent the suction pipe or pump; check for proper inlet conditions
Reduced or unstable discharge pressure	Leakage in suction pipe	Inspect and repair the suction pipe joints
Pump cavitates / loses prime	Pump inlet pressure too low	Check suction lift, clean strainer, verify NPSH available exceeds NPSH required
Seal weeping or visible leakage	Seal nearing end of service life	Schedule seal replacement per Section 1 interval